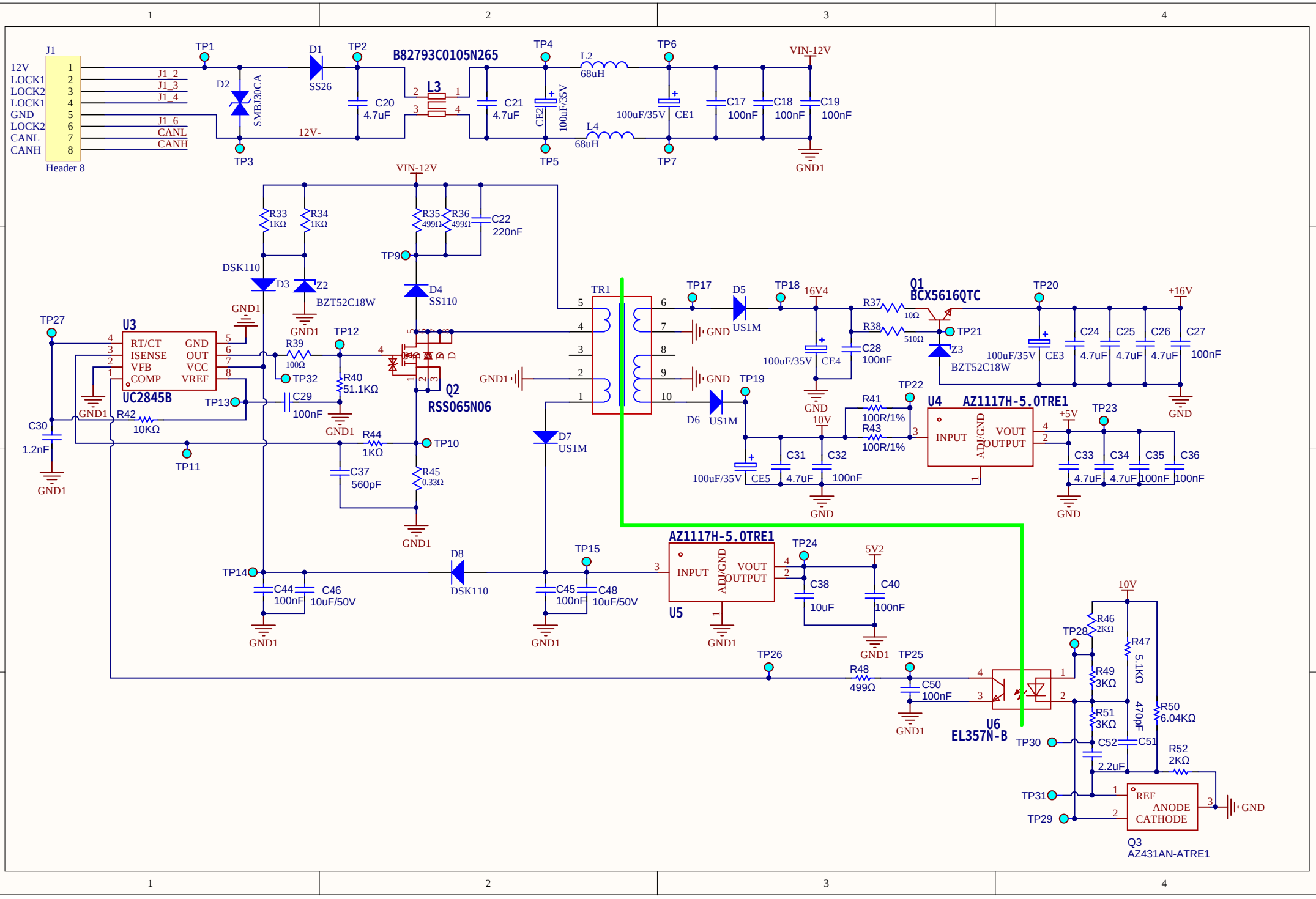
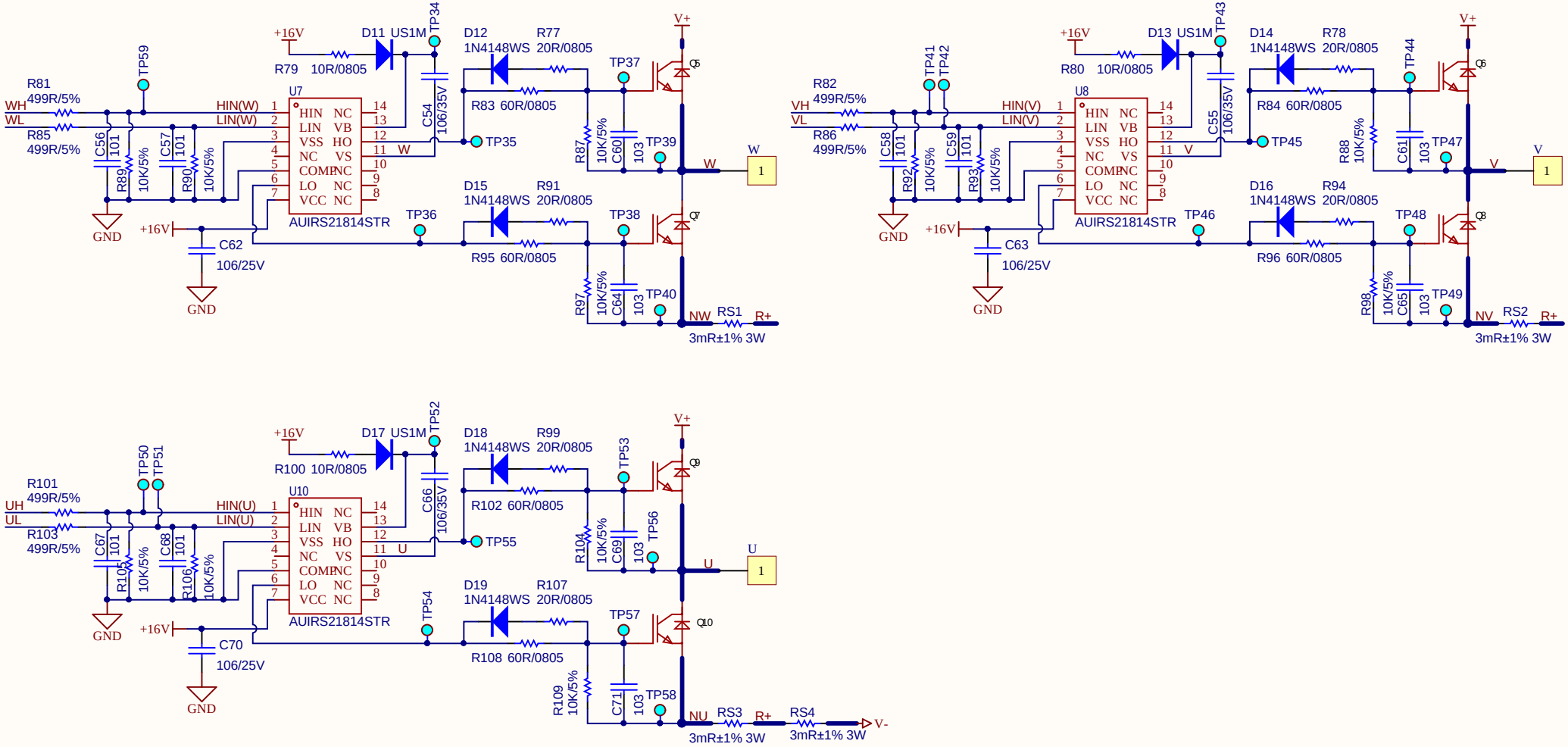
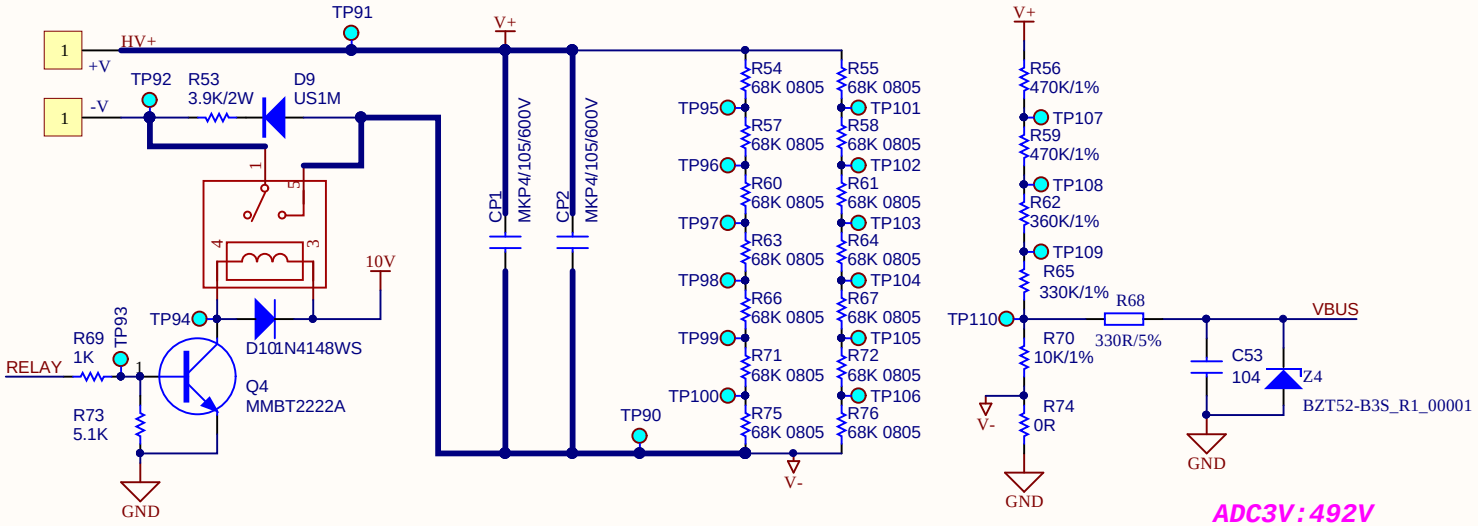

J1
12V
LOCK1
LOCK2
LOCK1
GND
LOCK2
CANL
CANH
Header 8
J1_2
J1_3
J1_4
J1_6
CANL
CANH
TP1
TP2
TP3
D1
SS26
D2
SMBJ30CA
12V-
B82793C0105N265
L3
C20
4.7uF
C21
4.7uF
TP4
TP5
CE2
100uF/35V
L2
68uH
L4
68uH
TP6
TP7
100uF/35V
CE1
C17
100nF
C18
100nF
C19
100nF
VIN-12V
GND1
VIN-12V
R33
1KΩ
R34
1KΩ
R35
499Ω
R36
499Ω
C22
220nF
TP9
DSK110
D3
Z2
BZT52C18W
D4
SS110
GND1
TR1
TP17
D5
US1M
TP18
16V4
Q1
BCX5616QTC
R37
10Ω
R38
510Ω
TP21
Z3
BZT52C18W
TP20
+16V
100uF/35V
CE3
C24
4.7uF
C25
4.7uF
C26
4.7uF
C27
100nF
GND
TP27
U3
RT/CT
ISENSE
VFB
COMP
GND
OUT
VCC
VREF
UC2845B
R39
100Ω
TP32
TP12
R40
51.1KΩ
C29
100nF
TP13
R42
10KΩ
C30
1.2nF
TP11
R44
1KΩ
C37
560pF
R45
0.33Ω
TP10
Q2
RSS065N06
GND1
D7
US1M
TP19
D6
US1M
GND
10V
100uF/35V
CE4
C28
100nF
TP22
R41
100R/1%
R43
100R/1%
U4
AZ1117H-5.0TRE1
INPUT
ADJ/GND
VOUT
OUTPUT
+5V
TP23
C31
4.7uF
C32
100nF
C33
4.7uF
C34
4.7uF
C35
100nF
C36
100nF
CE5
100uF/35V
TP14
D8
DSK110
C44
100nF
C46
10uF/50V
TP15
C45
100nF
C48
10uF/50V
AZ1117H-5.0TRE1
U5
TP24
5V2
C38
10uF
C40
100nF
TP26
R48
499Ω
TP25
C50
100nF
U6
EL357N-B
10V
TP28
R46
2KΩ
R47
5.1KΩ
R49
3KΩ
R50
6.04KΩ
R51
3KΩ
470pF
C52
C51
2.2uF
R52
2KΩ
TP30
TP31
TP29
REF
ANODE
CATHODE
Q3
AZ431AN-ATRE1

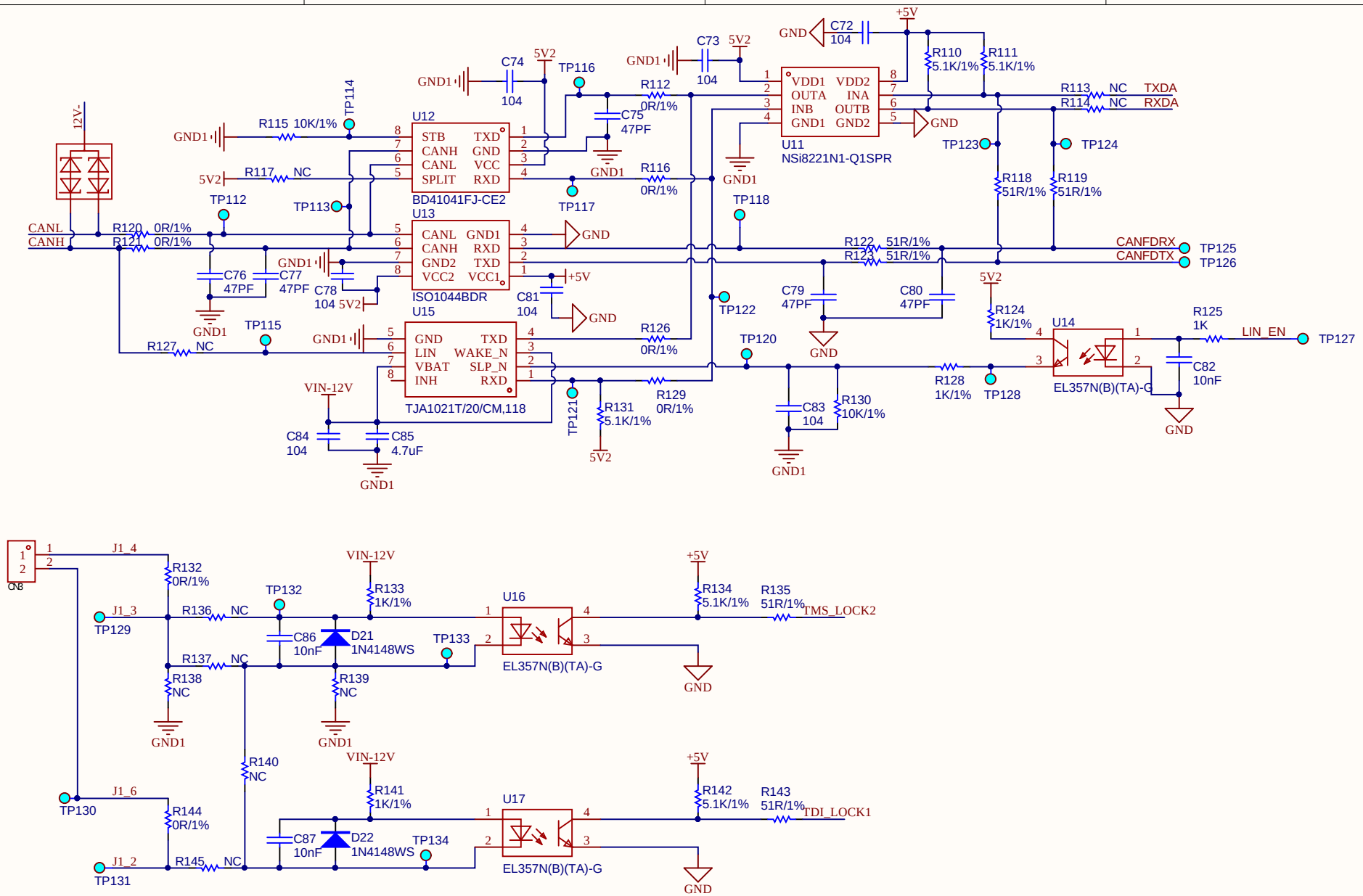

+16V
D11 US1M
TP34
D12
1N4148WS
R77
20R/0805
R79 10R/0805
R81
499R/5%
WH
WL
R85
499R/5%
TP59
HIN(W)
LIN(W)
U7
HIN NC
LIN VB
VSS HO
NC VS
COM NC
LO NC
VCC NC
AUIRS21814STR
C56
101
R89
10K/5%
C57
101
R90
10K/5%
C54
106/35V
R83 60R/0805
TP35
TP37
R87
10K/5%
C60
103
TP39
V+
W
1
D15
1N4148WS
R91
20R/0805
R95 60R/0805
TP36
TP38
R97
10K/5%
C64
103
TP40
NW
RS1
R+
3mR±1% 3W
GND
C62
106/25V
+16V
D13 US1M
TP43
D14
1N4148WS
R78
20R/0805
R80 10R/0805
R82
499R/5%
VH
VL
R86
499R/5%
TP41
TP42
HIN(V)
LIN(V)
U8
AUIRS21814STR
C58
101
R92
10K/5%
C59
101
R93
10K/5%
C55
106/35V
R84 60R/0805
TP45
TP44
TP47
R88
10K/5%
C61
103
V
D16
1N4148WS
R94
20R/0805
R96 60R/0805
TP46
TP48
R98
10K/5%
C65
103
TP49
NV
RS2
3mR±1% 3W
C63
106/25V
+16V
D17 US1M
TP52
D18
1N4148WS
R99
20R/0805
R100 10R/0805
R101
499R/5%
UH
UL
R103
499R/5%
TP50
TP51
HIN(U)
LIN(U)
U10
AUIRS21814STR
C67
101
R105
10K/5%
C68
101
R106
10K/5%
C66
106/35V
R102 60R/0805
TP55
TP53
R104
10K/5%
C69
103
TP56
U
D19
1N4148WS
R107
20R/0805
R108 60R/0805
TP54
TP57
R109
10K/5%
C71
103
TP58
NU
RS3
RS4
V-
C70
106/25V

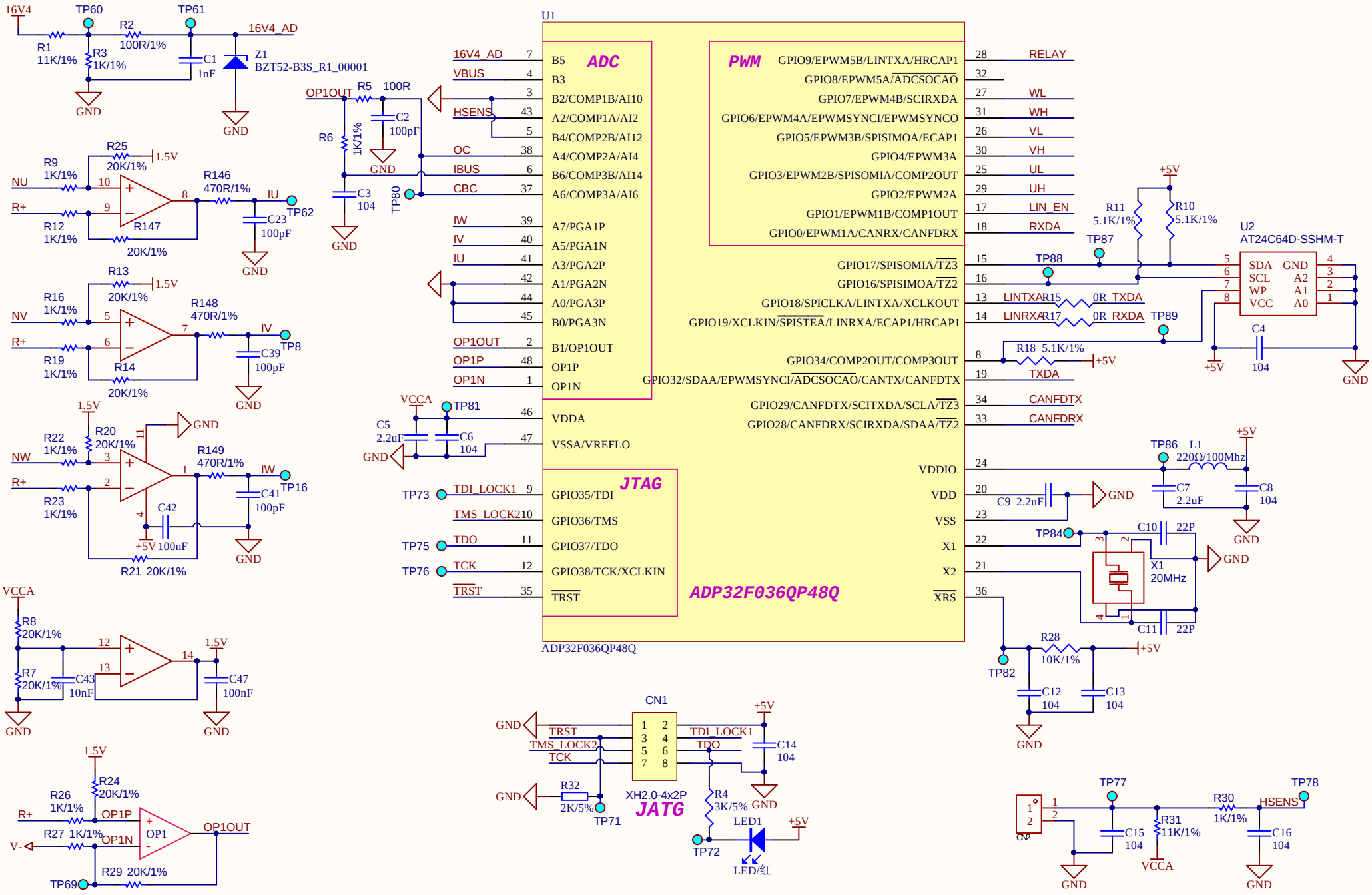


MARK 1 MARK 1 MARK 1

MARK 1 MARK 1 MARK 1

Pb


12V-
TP114
GND1
R115 10K/1%
5V2
R117 NC
TP112
TP113
CANL
CANH
R120 0R/1%
R121 0R/1%
C76 47PF
C77 47PF
GND1
C78 104
5V2
TP115
R127 NC
U12
STB TXD
CANH GND
CANL VCC
SPLIT RXD
BD41041FJ-CE2
U13
CANL GND1
CANH RXD
GND2 TXD
VCC2 VCC1
ISO1044BDR
U15
GND TXD
LIN WAKE_N
VBAT SLP_N
INH RXD
TJA1021T/20/CM,118
C74 104
5V2
TP116
TP117
C75 47PF
R112 0R/1%
R116 0R/1%
C81 104
+5V
GND
R126 0R/1%
VIN-12V
C84 104
C85 4.7uF
GND1
TP121
R131 5.1K/1%
5V2
R129 0R/1%
C73 104
5V2
GND1
U11
NSi8221N1-Q1SPR
VDD1 VDD2
OUTA INA
INB OUTB
GND1 GND2
GND
C72 104
+5V
R110 5.1K/1%
R111 5.1K/1%
R113 NC TXDA
R114 NC RXDA
TP123
TP124
R118 51R/1%
R119 51R/1%
TP118
TP122
R122 51R/1%
R123 51R/1%
CANFDRX
CANFDTX
TP125
TP126
C79 47PF
C80 47PF
5V2
R124 1K/1%
TP120
GND
R128 1K/1%
TP128
U14
EL357N(B)(TA)-G
R125 1K
LIN_EN
TP127
C82 10nF
GND
C83 104
R130 10K/1%
GND1
J1_4
R132 0R/1%
VIN-12V
TP132
R133 1K/1%
U16
+5V
R134 5.1K/1%
R135 51R/1%
TMS_LOCK2
J1_3
TP129
R136 NC
C86 10nF
D21 1N4148WS
TP133
EL357N(B)(TA)-G
GND
R137 NC
R138 NC
R139 NC
GND1
R140 NC
J1_6
TP130
R144 0R/1%
VIN-12V
R141 1K/1%
U17
+5V
R142 5.1K/1%
R143 51R/1%
TDI_LOCK1
C87 10nF
D22 1N4148WS
TP134
EL357N(B)(TA)-G
GND
J1_2
TP131
R145 NC


16V4
TP60
TP61
R1 11K/1%
R3 1K/1%
R2 100R/1%
C1 1nF
16V4_AD
Z1 BZT52-B3S_R1_00001
GND
OP1OUT
R5 100R
C2 100pF
R6 1K/1%
C3 104
TP80
U1
ADC
B5
B3
B2/COMP1B/AI10
A2/COMP1A/AI2
B4/COMP2B/AI12
A4/COMP2A/AI4
B6/COMP3B/AI14
A6/COMP3A/AI6
A7/PGA1P
A5/PGA1N
A3/PGA2P
A1/PGA2N
A0/PGA3P
B0/PGA3N
B1/OP1OUT
OP1P
OP1N
VDDA
VSSA/VREFLO
JTAG
GPIO35/TDI
GPIO36/TMS
GPIO37/TDO
GPIO38/TCK/XCLKIN
TRST
PWM
GPIO9/EPWM5B/LINTXA/HRCAP1
GPIO8/EPWM5A/ADCSOCAO
GPIO7/EPWM4B/SCIRXDA
GPIO6/EPWM4A/EPWMSYNCI/EPWMSYNCO
GPIO5/EPWM3B/SPISIMOA/ECAP1
GPIO4/EPWM3A
GPIO3/EPWM2B/SPISOMIA/COMP2OUT
GPIO2/EPWM2A
GPIO1/EPWM1B/COMP1OUT
GPIO0/EPWM1A/CANRX/CANFDRX
GPIO17/SPISOMIA/TZ3
GPIO16/SPISIMOA/TZ2
GPIO18/SPICLKA/LINTXA/XCLKOUT
GPIO19/XCLKIN/SPISTEA/LINRXA/ECAP1/HRCAP1
GPIO34/COMP2OUT/COMP3OUT
GPIO32/SDAA/EPWMSYNCI/ADCSOCAO/CANTX/CANFDTX
GPIO29/CANFDTX/SCITXDA/SCLA/TZ3
GPIO28/CANFDRX/SCIRXDA/SDAA/TZ2
VDDIO
VDD
VSS
X1
X2
XRS
ADP32F036QP48Q
16V4_AD
VBUS
HSENS
OC
IBUS
CBC
IW
IV
IU
OP1OUT
OP1P
OP1N
VCCA
TP81
C5 2.2uF
C6 104
TP73
TDI_LOCK1
TMS_LOCK2
TP75
TDO
TP76
TCK
TRST
RELAY
WL
WH
VL
VH
UL
UH
LIN_EN
RXDA
TP88
LINTXA
LINRXA
R15 0R_TXDA
R17 0R_RXDA
TP89
R18 5.1K/1%
TXDA
CANFDTX
CANFDRX
+5V
R11 5.1K/1%
R10 5.1K/1%
TP87
U2
AT24C64D-SSHM-T
SDA
SCL
WP
VCC
GND
A2
A1
A0
C4 104
TP86
L1 220Ω/100Mhz
C7 2.2uF
C8 104
C9 2.2uF
TP84
C10 22P
X1 20MHz
C11 22P
R28 10K/1%
TP82
C12 104
C13 104
R9 1K/1%
R12 1K/1%
R25 20K/1%
R146 470R/1%
R147 20K/1%
C23 100pF
TP62
NU
R+
R13 20K/1%
R16 1K/1%
R19 1K/1%
R14 20K/1%
R148 470R/1%
C39 100pF
TP8
NV
R20 20K/1%
R22 1K/1%
R23 1K/1%
R149 470R/1%
C41 100pF
TP16
NW
C42 100nF
R21 20K/1%
R8 20K/1%
R7 20K/1%
C43 10nF
C47 100nF
R24 20K/1%
R26 1K/1%
R27 1K/1%
R29 20K/1%
OP1
TP69
V-
CN1
XH2.0-4x2P
JATG
R32 2K/5%
TP71
R4 3K/5%
LED1
TP72
LED/红
CN2
TP77
TP78
R30 1K/1%
R31 11K/1%
C15 104
C16 104
HSENS